



BITS Pilani
K K Birla Goa Campus

A Multi-Scale Analysis of Reliance Industries(2011-2025)

A Complexity Driven, Valuation Conditioned Framework

Rishabh Sahu (2022B5A71241G)

Course : PHY F366

Under the guidance of:

Prof. Chandradew Sharma

Department of Physics
Birla Institute of Technology and Science, Pilani

K.K Birla Goa Campus

November 15, 2025

Abstract

This project applies the multi-scale statistical, complexity-theoretic, and information-theoretic framework proposed in *Sharma (2024)* to a single-stock application using Reliance Industries Ltd. Monthly valuation and return data from January 2011 to January 2025 are used to estimate conditional return distributions across multiple investment horizons, characterize complexity through entropy, Hurst exponent, and Lyapunov exponent, and analyze nonlinear information flow via normalized mutual information (NMI) and transfer entropy (TE).

Results indicate that the valuation series (P/E TTM) exhibits persistent ($H \approx 0.95$) and chaotic ($LLE > 0$) behavior, while monthly returns display anti-persistence ($H \approx 0.36$). Conditional valuation-band analysis shows that lower P/E regimes historically correspond to higher positive-return probabilities, especially at multi-year horizons. Information-flow tests confirm $P/E \rightarrow \text{returns}$ as the dominant direction of nonlinear information transmission.

Overall, valuation plays a meaningful role in long-horizon prediction for Reliance Industries, consistent with the findings of the broader market analysis in *Sharma (2024)*.

1.Introduction

Forecasting the behavior of financial markets is challenging due to nonlinearity, noise, and structural instability. Traditional linear metrics often fail to explain long-run patterns in return behavior, particularly during extended valuation cycles.

The methodology introduced in *Sharma (2024)* integrates:

1. **Valuation Regime Conditioning**
 - Partitioning the valuation distribution into bands and computing outcome probabilities.
2. **Complexity Diagnostics**
 - Using entropy, Hurst exponent, and Lyapunov exponent to characterize persistence, memory, and chaos.
3. **Information-Theoretic Causality**
 - Using mutual information and transfer entropy to quantify nonlinear dependence and directional information flow.

While the paper focuses on the Nifty 50 index, this project adapts the methodology to a **single stock: Reliance Industries Ltd.**

The key research question:

Does valuation (P/E TTM) provide predictive information about future returns for Reliance Industries when analyzed through a multi-scale and complexity-aware framework?

Multi-Scale Analysis of Nifty 50: Statistical, Complexity, and Information-Theoretic Approach

The original paper, “*Multi-Scale Analysis of Nifty 50: Statistical, Complexity, and Information-Theoretic Approach*,” conducts a comprehensive multi-horizon evaluation of Nifty valuations, returns, and information flow. Its key contributions are:

A. Valuation Regimes and Conditional Forecasting

- The Nifty's P/E distribution is used to define valuation bands (e.g., <13 = low valuation).
- Higher future returns and lower downside risk are observed during low valuation regimes.
- Reward-risk ratios improve dramatically with horizon length.

B. Complexity Diagnostics

- Hurst exponents > 0.5 for valuations $\rightarrow$ persistent, trend-following valuation cycles.
- Returns show mean-reversion, not persistence.
- Positive Lyapunov exponents for valuation $\rightarrow$ chaotic but structured long-term patterns.

C. Information Flow and Causality

- Transfer Entropy shows valuation $\rightarrow$ returns as the dominant direction.
- Suggests valuation acts as a *leading indicator*.

D. Practical Implication

- Valuation-based regime conditioning significantly improves long-horizon forecasting accuracy

This study applies the same lens to Reliance.

2.Data Sources and Preprocessing

2.1 Data Sources

Monthly Price Data

Downloaded from:

Investing.com – Reliance Industries Historical Prices

<https://in.investing.com/equities/reliance-industries-historical-data>

Quarterly EPS Data

Downloaded from:

Macrotrends – Reliance EPS (Diluted, Quarterly)

<https://www.macrotrends.net/stocks/charts/RS/reliance/eps-earnings-per-share-diluted>

2.2 Construction of Monthly Trailing-Twelve-Month (TTM) P/E Ratio

Since EPS is reported quarterly and price is available monthly, the P/E time series was constructed as follows:

Step 1: Compute Quarterly TTM EPS

For every quarterly EPS observation, the TTM EPS is computed as:

$$TTM\ EPS_t = EPS_t + EPS_{t-1} + EPS_{t-2} + EPS_{t-3}$$

This provides the earnings accumulated over the last four quarters.

Step 2: Forward-Fill Monthly TTM EPS

Because earnings are reported only quarterly, the latest known TTM EPS is **forward-filled** to each subsequent monthly price until a new EPS release occurs.

This ensures each month has a corresponding TTM EPS value.

Step 3: Compute Monthly P/E Ratio

For each monthly closing price

$$P/E_m = \frac{Price_m}{TTM\ EPS_{latest}}$$

Where

TTM EPS_{latest} = most recently available quarterly TTM EPS at that month.

This method replicates valuation construction used by equity researchers, Bloomberg, and datasets like Nifty valuations.

2.3 Final Dataset Summary

- **Time span:** Jan 2011 – Jan 2025
- **Observations:** 168 monthly P/E samples
- **Derived variables:**
 - Monthly returns
 - 1-month / 12-month P/E changes
 - 1–7 year forward CAGRs
 - Valuation quantiles (Q1, Q2, Q3)

Valuation bands used in the analysis:

Band	P/E Range
Q1 (Low)	< 43.49
Q2 (Mid-Low)	43.5 – 52.1
Q3 (Mid-High)	52.1 – 71.2
Q4 (High)	> 71.2

3.Statistical Return and Valuation Analysis

P/E Distribution Statistics

- Mean P/E: **61.43**
- Median P/E: **52.11**
- Std. Dev.: **27.34**
- Mode: **39.86**

Return Sensitivity to P/E Change

1-month results:

- PRP: **52.1%**
- RRR: **1.23**

12-month results:

- PRP: **64.1%**
- RRR: **1.39**

Short-term P/E changes have limited predictive power; medium-term changes show clearer trends.

4.Complexity Analysis

Using `nolds` and entropy estimation:

4.1 P/E Ratio Complexity

- **Shannon Entropy:** 0.805
- **Hurst Exponent:** 0.947
- **Lyapunov Exponent:** +0.014

Interpretation:

- *Persistent:* P/E deviations tend to continue.
- *Chaotic:* Short-term valuation movements are sensitive and unpredictable.
- Indicates structural valuation regimes (matching Sharma, 2024 findings for Nifty valuations).

4.2 Price Return Complexity

- **Shannon Entropy:** 0.843
- **Hurst Exponent:** 0.356
- **Lyapunov Exponent:** -0.003

Interpretation:

- Monthly returns are **anti-persistent** (mean-reverting).
- No strong evidence of chaos.
- Price changes behave closer to a random process than valuation.

5. Conditional P/E Band Analysis

Key findings from your computed table:					
P/E Band	1-Year PRP	2-Year PRP	3-Year PRP	5-Year PRP	RRR
Low (Q1)	0.86	0.98	1.00	1.00	Very High
Mid-Low (Q2)	0.64	0.97	1.00	1.00	Strong ↑
Mid-High (Q3)	0.72	0.91	1.00	1.00	Moderate
High (Q4)	0.81	1.00	1.00	1.00	Mixed

6. Information Flow Analysis

6.1 Normalized Mutual Information (NMI)

- NMI = 0.1253

This is high for financial data and indicates **significant nonlinear dependence**.

6.2 Transfer Entropy (TE)

- TE(P/E → Returns) = 0.5402
- TE(Returns → P/E) = 0.2099

Interpretation:

- Information flow from valuation **to** returns is **157% stronger** than the reverse.
- This supports valuation as a **leading variable**, consistent with Sharma (2024).

7. Interpretations

7.1 Overall interpretations

1. **Valuation is persistent and chaotic, forming meaningful regimes.**
2. **Returns are mean-reverting**, reducing short-term predictability.
3. **Low P/E bands yield superior long-term outcomes.**
4. **P/E → Returns** is the dominant direction of information flow.
5. Reliance behaves similarly to the index-level patterns documented in the reference paper.

7.2 Practical Implications

For Investors

- Favor allocation increases during **Low P/E (< 43.5)** periods.
- Exercise caution in **High P/E (> 71.2)** regimes.
- Use valuation conditioning for **1–7 year horizons**, not short-term timing.

For Quant Researchers

- Combine P/E with earnings momentum for improved forecasts.
- Use regime-based backtesting rather than linear models.
- Recompute quantiles periodically to detect structural shifts.

8. Limitations

- Results may be affected by **single-stock noise** and **limited sample size**.
- Some RRR values approach infinity due to **zero negative outcomes** in certain bins.
- Chaotic valuation implies **limited short-term predictability**.
- EPS revisions or changes in accounting standards can alter TTM EPS structure.

9. Conclusion

This study successfully reproduces the analysis framework of *Sharma (2024)* on a single-stock basis for Reliance Industries. Despite the inherent noise of individual equities, the core findings mirror those observed at the index level:

- Valuation (P/E TTM) is persistent, nonlinear, and chaotic.
- Returns are mean-reverting and less predictable short-term.
- Conditional valuation bands capture meaningful differences in long-horizon outcomes.
- Transfer entropy validates valuation as a directional predictor.

Thus, the P/E ratio—constructed carefully using quarterly TTM EPS forward-filled to monthly frequency—provides a robust conditioning tool for long-horizon investment decisions in Reliance Industries.

10. References

1. Sharma, Chandradew. *Multi-Scale Analysis of Nifty 50: Statistical, Complexity, and Information-Theoretic Approach*, SSRN (2024).
2. Monthly price data sourced from Investing.com:
<https://in.investing.com/equities/reliance-industries-historical-data>
3. Quarterly EPS data sourced from Macrotrends:
<https://www.macrotrends.net/stocks/charts/RS/reliance/eps-earnings-per-share-diluted>
4. Complexity computation libraries: `nolds`, `scipy`, `numpy`.
5. Transfer entropy estimation via JIDT-inspired Python implementation.